CL and NLP
Assignment - 1

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Ques1.

$x_t = 0.5$

$h_{t-1} = 0.2$

$C_{t-1} = 0.4$

LSTM equations are:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

$$\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$

$$h_t = o_t \tanh(C_t)$$

a) Forget gate:

$$z_f = (0.7)(0.5) + (0.3)(0.2) + 0.1 = 0.5100$$

$$\mathbf{f_t = \sigma(z_f) = 1 / (1 + e^{-0.5100}) = 0.6248}$$

b) Input gate:

$$z_i = (0.6)(0.5) + (0.4)(0.2) + 0 = 0.3800$$

$$\mathbf{i_t = 1 / (1 + e^{-0.3800}) = 0.5939}$$

c) Candidate cell state ($\tilde{C}_t$):

$$z_c = (0.9)(0.5) + (0.2)(0.2) = 0.4900$$

$$\mathbf{\tilde{C}_t = \tanh(0.4900) = 0.4542}$$

d) New cell state (C_t):

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

$$\mathbf{C_t = (0.6248)(0.4) + (0.5939)(0.4542) = 0.5197}$$

e) Output gate (o_t):

$$z_o = (0.5)(0.5) + (0.5)(0.2) + 0.1 = 0.4500$$

$$\mathbf{o_t = 1 / (1 + e^{-0.4500}) = 0.6106}$$

f) New hidden state (h_t):

$$h_t = o_t \tanh(C_t)$$

$$\tanh(0.5197) \approx 0.4775$$

$$h_{\square} = (0.6106)(0.4775) \approx 0.2915$$

Ques2.

D = 10

H = 32

Number of gates = 4

a) Parameters:

$$P = 4(DH + H^2 + H)$$

$$P = 4[(10)(32) + (32)(32) + 32]$$

$$\mathbf{P = 5504 \text{ parameters}}$$

(one bias vector per gate)

b)

$$P = 4(DH + H^2 + 2H)$$

$$P = 4[(10)(32) + (32)(32) + 2(32)]$$

$$\mathbf{P = 5632 \text{ parameters}}$$

(two bias vectors per gate)

Ques**3.**

$$\textbf{Scaling factor} = f^k$$

a) f = 0.9

After 10 steps:

$$\mathbf{f^{10} = (0.9)^{10} = 0.3487}$$

After 50 steps:

$$\mathbf{f^{50} = (0.9)^{50} = 0.0052}$$

b) For k = 50:

When f = 0.5:

$$\mathbf{f^{50} = (0.5)^{50} = 8.8818 \times 10^{-16}}$$

When f = 0.99:

$$\mathbf{f^{50} = (0.99)^{50} = 0.6050}$$

c)

When the forget gate is 0.5, the gradient becomes almost zero after 50 time steps, making it extremely difficult for the network to learn long-term dependencies. In contrast, when the forget gate is 0.99, approximately 60.50% of the gradient remains after 50 steps.

Ques4. Given:

Duration: 3 years

Days per year: 365

Hours per day: 24

Input lookback: 168 hours

Forecast horizon: 24 hours

Features per hour: 3

Stride: 1 hour

Batch size: 64

a) Total number of hourly records:

$$\mathbf{N = 3 \times 365 \times 24 = 26280}$$

b) For stride = 1:

$$\text{Windows} = N - L - H + 1$$

$$\mathbf{Windows = 26280 - 168 - 24 + 1 = 26089}$$

c) For batch of 64 windows:

$$\mathbf{X: (64, 168, 3)}$$

Each target contains next 24 hourly load values.

$$\mathbf{Y: (64, 24)}$$

d)

$$\mathbf{Ntrain = 0.70 \times 26280 = 18396}$$

$$\mathbf{Wtrain = 18396 - 168 - 24 + 1 = 18205}$$

$$\text{Batches} = \text{Training windows} / \text{Batch size} = 18205 / 64$$

$$\mathbf{= \lceil 284.4531 \rceil = 285 \text{ mini-batches per epoch}}$$

e)

$$P = 4(DH + H^2 + H)$$

$$\mathbf{P = 4[(3)(64) + (64)^2 + 64] = 17408}$$

5.

a) Min-Max normalisation:

$$x' = (x - x_{min}) / (x_{max} - x_{min})$$

$$\mathbf{x' = (95 - 20) / (120 - 20) = 0.75}$$

b)

Prediction = Failure, if $p \geq 0.5$

Prediction = No failure, if $p < 0.5$

$$0.83 \geq 0.5$$

Prediction: Failure

c)

True positives (TP) = 30

False positives (FP) = Predicted failures - TP

= 45 - 30 = 15

FP = 15

False negatives (FN) = Actual failures - TP

= 40 - 30 = 10

FN = 10

Actual non-failures = 1000 - 40 = 960

TN = Actual non-failures - FP

= 960 - 15 = 945

TN = 945

Precision:

TP / (TP + FP) = 30 / (30 + 15) = 66.67%

Recall:

TP / (TP + FN) = 30 / (30 + 10) = 75%

F1-score:

2 × Precision × Recall / (Precision + Recall)

F1-score = [2 × (30/45) × (30/40)] / [(30/45) + (30/40)] = 70.59%

Accuracy:

(TP + TN) / (TP + TN + FP + FN)

(30 + 945) / 1000 = 0.9750 = 97.50%

d)

It should generally prioritize recall, because failing to detect a genuinely failing pump can cause equipment damage, unexpected downtime, or safety risks.

Accuracy can be misleading because the dataset is imbalanced: only 40/1000 samples represent failures. A model that predicts “no failure” for every case would still achieve 96% accuracy while detecting none of the failures. Therefore, recall, together with precision and F1-score, provides a more useful assessment of failure detection.